2. What is hoisting and what is the Temporal Dead Zone (TDZ)? Explain with examples.

Hoisting is a JavaScript behavior where variable and function declarations are processed before the code is executed.
When using var, the variable is hoisted and initialized with undefined. This means you can access it before its declaration,but its value will be undefined.
Example:

```
console.log(x);
var x = 10;
```

Output:
Undefined

With let and const, the variables are also hoisted, but they are not initialized immediately. The time between entering the scope and declaring the variable is called the Temporal Dead Zone (TDZ).If you try to access the variable during this period, JavaScript throws a ReferenceError.
Example:

```
console.log(age);
let age = 20;
```

Output:
ReferenceError

* The TDZ helps prevent using variables before they are properly initialized, making the code safer and easier to understand.

3. What are the main differences between == and ===?

The "==" operator compares values after performing **type coercion**, which means JavaScript may automatically convert one data type to another before comparing.

Example:
5 == "5"
Output:
True

The === operator is called the strict equality operator. It compares both the value and the data type without converting anything.

Example:
5 === ″5″
Output:
False

4. What’s the difference between type conversion and coercion? Provide examples of each.

Both type conversion and type coercion change the data type of a value, but the difference is who performs the conversion.

* Type conversion (explicit conversion) happens when the programmer converts the data type manually using functions like Number(), String(), or Boolean().
Example:

let num = Number(″25″);

Output:
25

* Type coercion (implicit conversion) happens automatically by JavaScript when different data types are used together.

Example:

″5″ + 2

Output:
″52″

Another example:

″5″ - 2

Output: